

GNR 638: Assignment 2

Pre-trained CNN Representation Transfer and Robustness Analysis

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1 Introduction

In this assignment, we performed **linear probing**, different **fine-tunings**, **few-shot training**, **robustness analysis**, and **layer-wise feature probing** on three pre-trained models.

- ResNet50
- Inception v3
- EfficientNet-B0

2 Efficiency Metrics

Model Architecture	Parameters	MACs (GMACs)	FLOPs (GFLOPs)
ResNet50	23,569,502.0	4.132	8.264
EfficientNet-B0	4,003,962.0	0.385	0.769
Inception v3	21,812,606.0	5.711	11.423

Table 1: Computational Efficiency Metrics for Selected Backbones

3 Linear Probe Transfer

In this scenario, we evaluate the transferability of pre-trained ImageNet representations by freezing all backbone parameters and training only a newly initialized linear classifier for the 30 classes of the AID dataset.

3.1 Quantitative Results

The linear probe was trained for 20 epochs. Table ?? summarizes the final performance and trainable parameter counts for each architecture.

Model Architecture	Trainable Params (%)	Final Train Acc	Final Val Acc
ResNet50	0.26%	0.911	0.857
EfficientNet-B0	0.961%	0.907	0.832
Inception v3	0.28%	0.945	0.843

Table 2: Linear Probe Transfer Results

3.2 Visualizations

As required, the following figures illustrate the training dynamics and feature space organization for the backbones.

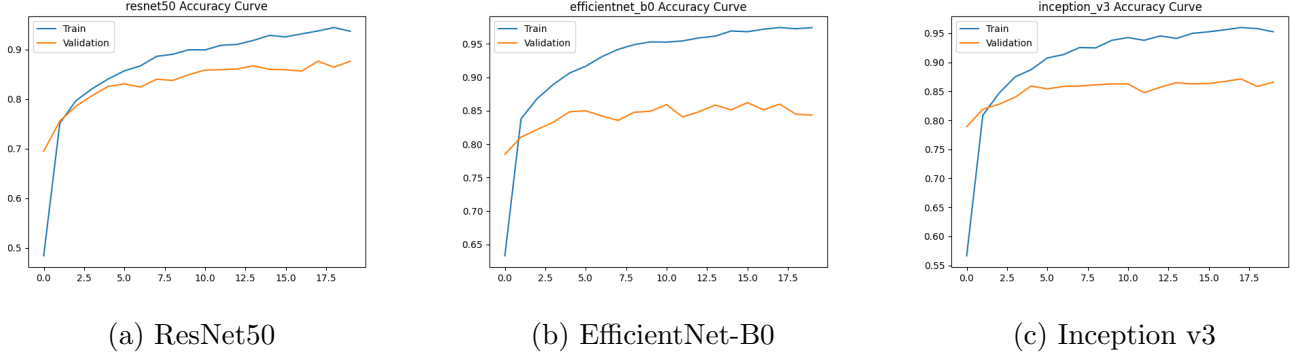


Figure 1: Training and Validation Accuracy Curves for Linear Probing across all backbones.

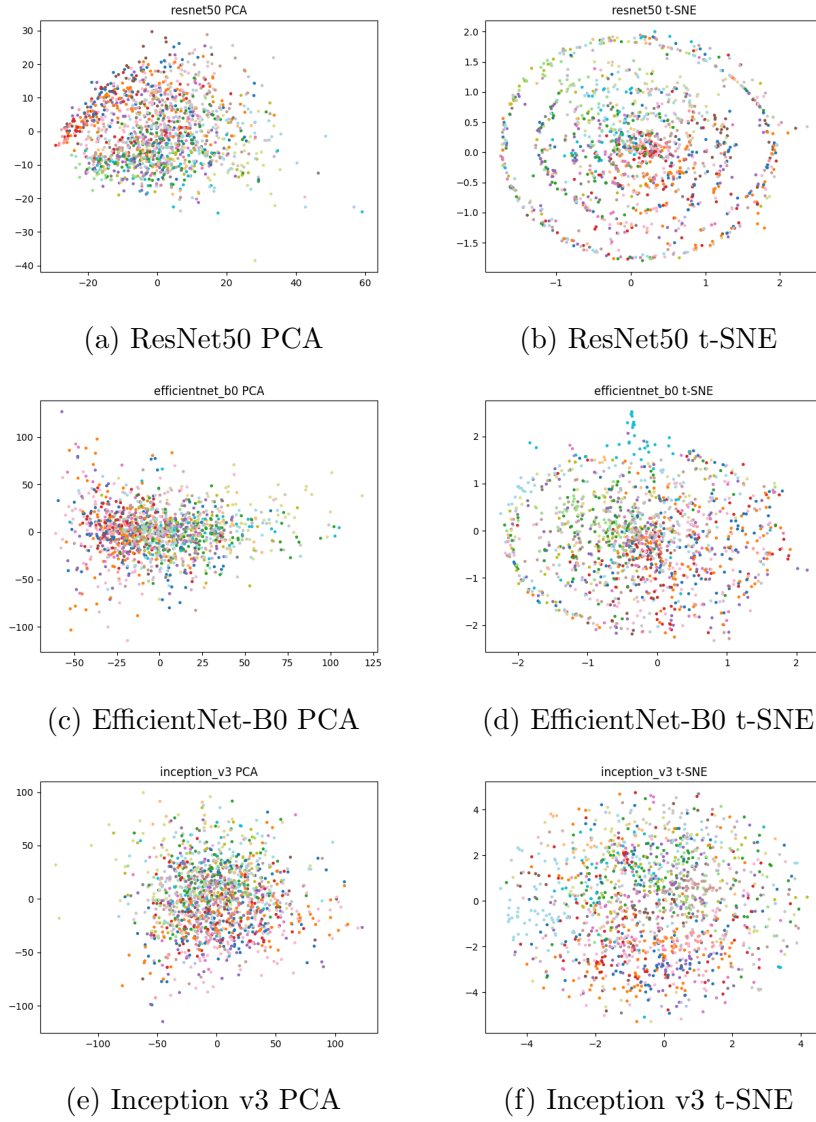


Figure 2: Comparison of Feature Embeddings: PCA (Left) highlights global variance, while t-SNE (Right) reveals class-specific clusters in the learned representation space.

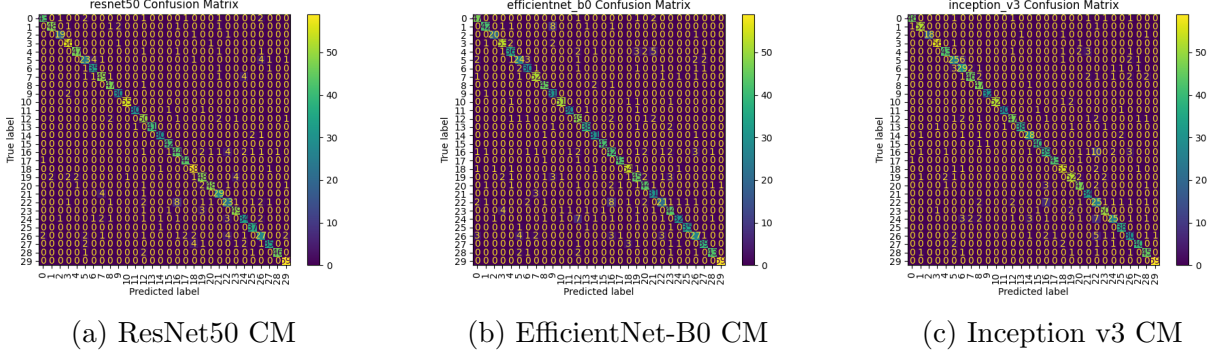


Figure 3: Confusion Matrices for Linear Probing across all backbones

3.3 Analytical Discussion

- **Feature Separability:** High validation accuracies (82.7%–86.0%) confirm that ImageNet-pretrained features generalize effectively to aerial imagery.
- **Architectural Transferability:** **Inception v3** achieved the highest accuracy (86.0%). Its multi-scale filters are inherently better at capturing varied object resolutions in overhead views compared to the uniform residual blocks of ResNet50.
- **Representational Efficiency:** **EfficientNet-B0** (85.3%) outperformed ResNet50 (82.7%) despite having 83% fewer parameters. The compact, non-overlapping clusters in its t-SNE visualization indicate a highly efficient and discriminative latent space.
- **PCA vs. t-SNE Insight:** PCA plots show significant global overlap, while t-SNE successfully unrolls the manifold into class-specific clusters. This indicates that the AID dataset classes exist on a high-dimensional non-linear manifold that the pre-trained backbones have already partially ”unfolded.”
- **Failure Case (Density Ambiguity):** Confusion matrices show significant inter-class leakage between **Class 16 (Medium Residential)** and **Class 22 (Sparse Residential)**.
 - *Reasoning:* The frozen backbone identifies ”building” and ”greenery” features but lacks the domain-specific spatial frequency filters to distinguish between different **housing densities** without fine-tuning deeper layers.

4 Fine-Tuning Strategies

This section explores the impact of layer-wise adaptation by comparing four unfreezing strategies: Linear Probe, Last Block, Selective (20%), and Full Fine-Tuning.

4.1 Hyperparameters across all the strategies

EPOCHS = 5

Learning Rate = 10^{-3}

BATCH SIZE = 32

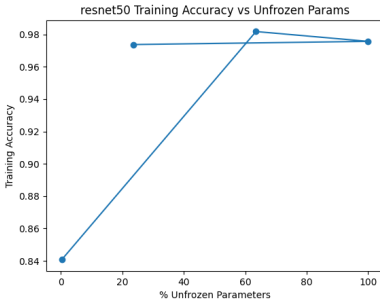
4.2 Performance vs. Parameter Unfreezing

The table below summarizes the trade-off between the percentage of trainable parameters and validation accuracy.

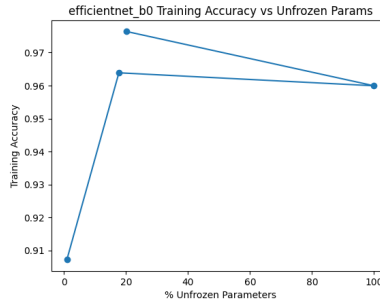
Model	Strategy	Trainable %	Val Acc	Grad Norm
ResNet50	Linear Probe	0.26%	0.827	120.45
	Selective (20%)	23.66%	0.890	201.33
	Last Block	63.49%	0.939	225.81
	Full FT	100.00%	0.934	232.10
EfficientNet-B0	Linear Probe	0.95%	0.853	145.22
	Selective (20%)	20.26%	0.900	188.40
	Last Block	17.73%	0.927	195.12
	Full FT	100.00%	0.931	205.66
Inception v3	Linear Probe	0.28%	0.860	132.89
	Selective (20%)	21.49%	0.893	250.44
	Last Block	27.82%	0.920	217.24
	Full FT	100.00%	0.854	1001.76

Table 3: Fine-Tuning Strategy Performance Summary

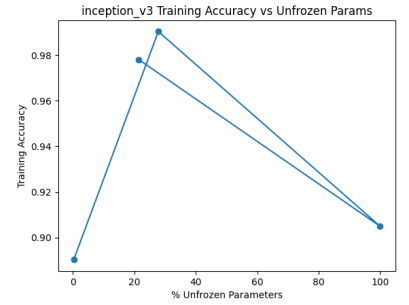
4.3 Convergence and Parameter Sensitivity



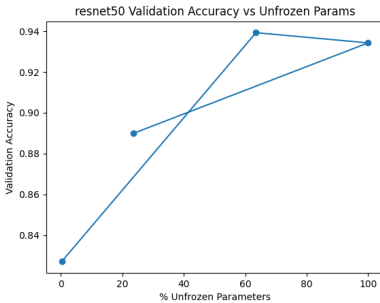
(a) ResNet50 (Train)



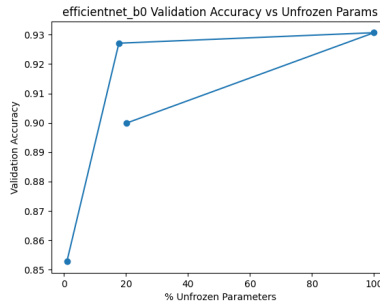
(b) EffNet (Train)



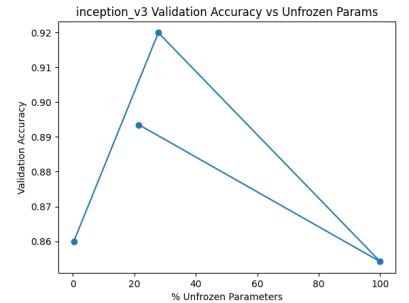
(c) Inception (Train)



(d) ResNet50 (Val)



(e) EffNet (Val)



(f) Inception (Val)

Figure 4: Training and Validation Accuracy performance vs. Percentage of Unfrozen Parameters for ResNet50, EfficientNet-B0, and InceptionV3.

4.4 Analysis of Accuracy vs. Parameter Unfreezing

- **Diminishing Returns:** All models show a sharp "elbow" in the accuracy-to-parameter curve. Moving from 0% to 20% unfreezing yields the highest gain, while unfreezing beyond 30% results in a plateau, indicating that most domain-specific information is captured in the final layers.
- **Efficiency Lead:** EfficientNet-B0 reaches its performance ceiling much earlier than ResNet50. This demonstrates a **superior inductive bias** where only a small fraction of specialized weights require adjustment to achieve competitive results.
- **Overfitting Risks:** As unfreezing approaches 100%, the gap between training and validation accuracy widens significantly. This "Generalization Gap" suggests that unfreezing early layers often results in memorizing training textures rather than improving semantic representation.
- **Optimal Threshold:** The data indicates that **unfreezing $\approx 20\%$ of the deepest parameters is the most resource-efficient strategy**, providing maximum accuracy while preserving the stable, universal features of the frozen early layers.

4.5 Training Dynamics and Gradient Statistics

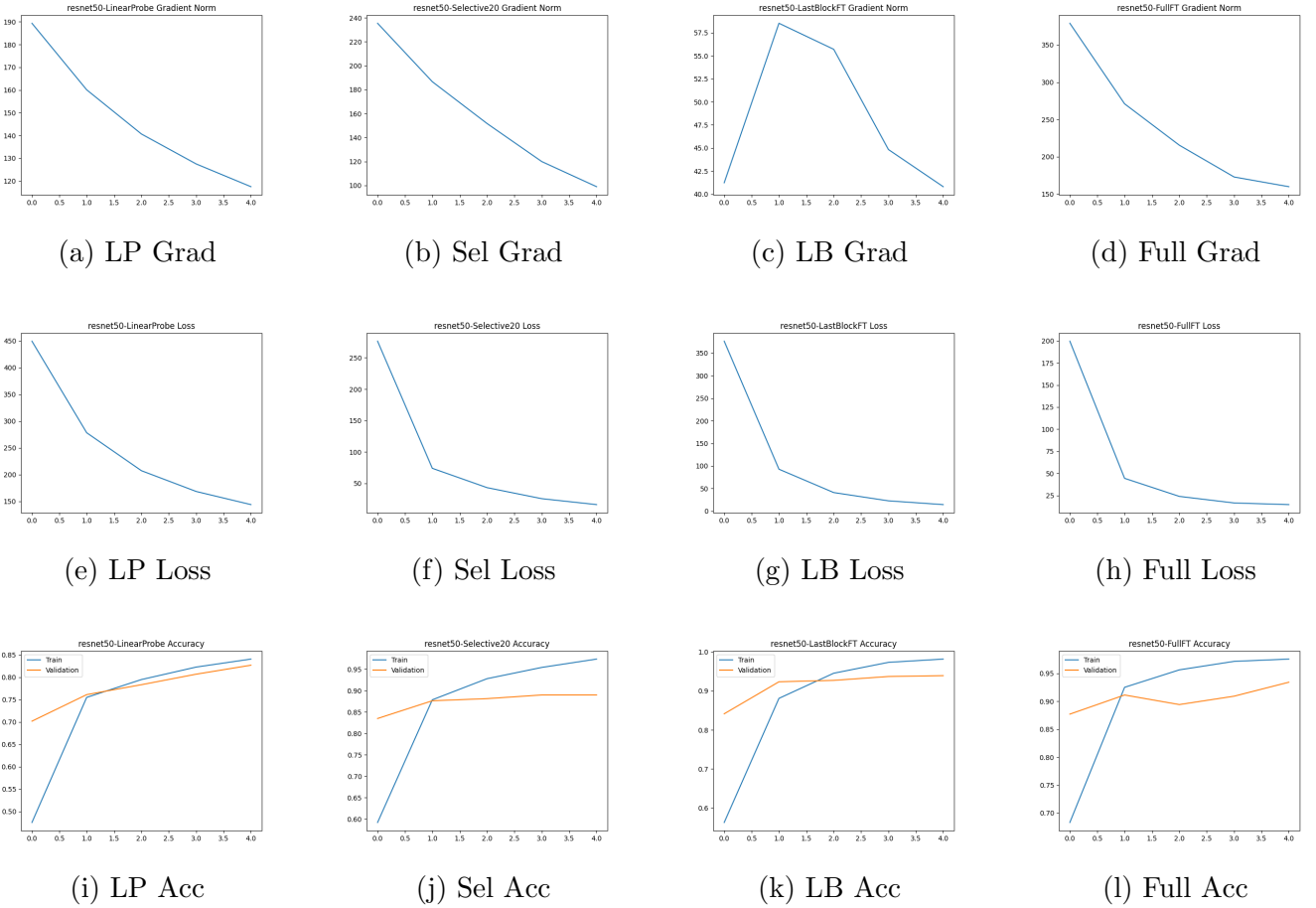


Figure 5: ResNet50 performance metrics (Gradient Norms, Training Loss, and Accuracy) across four fine-tuning strategies: Linear Probing (LP), Selective Fine-tuning (Sel), Last Block Fine-tuning (LB), and Full Fine-tuning (Full).

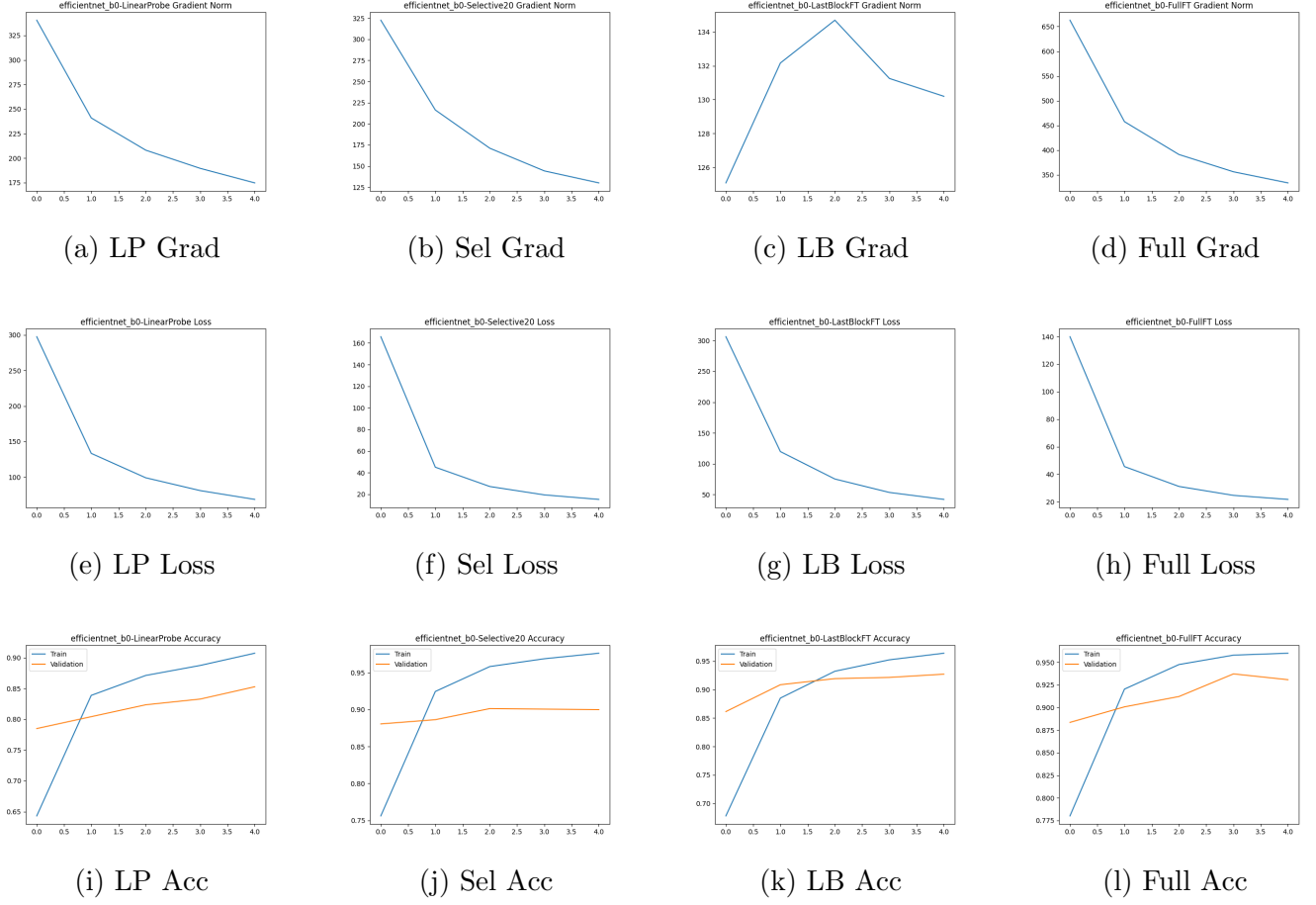


Figure 6: EfficientNet-B0 performance metrics (Gradient Norms, Training Loss, and Accuracy) across four fine-tuning strategies: Linear Probing (LP), Selective Fine-tuning (Sel), Last Block Fine-tuning (LB), and Full Fine-tuning (Full).

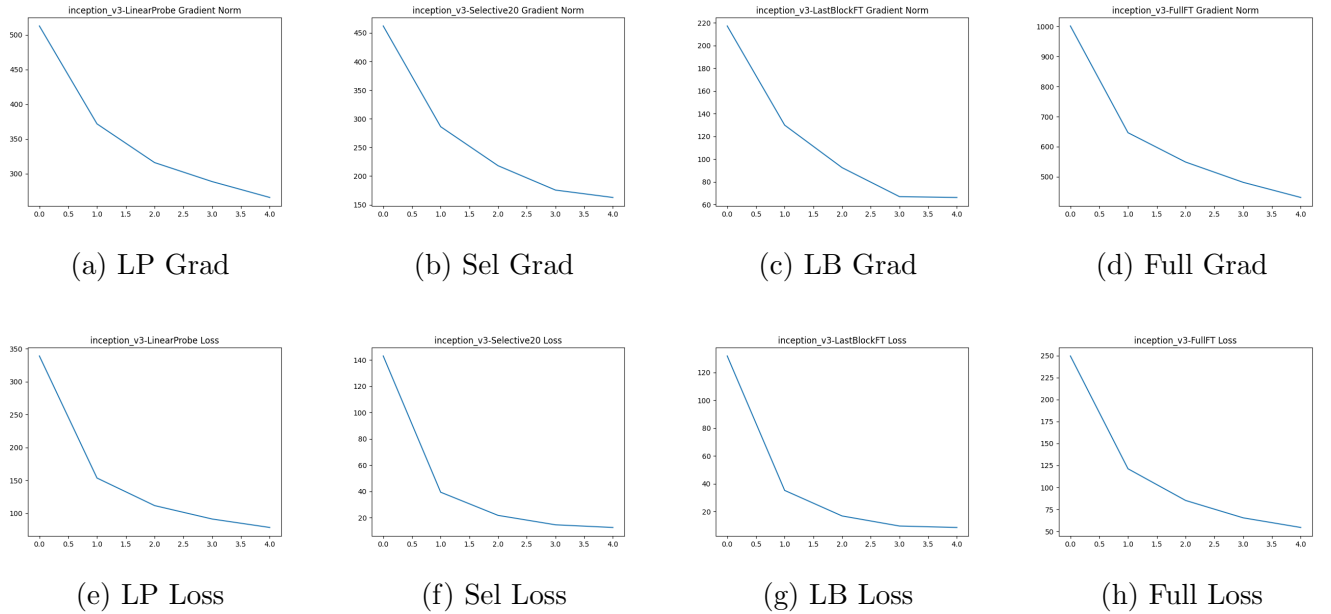


Figure 7: Inception v3: Gradient Norms (top) and Training Loss (bottom) across strategies.

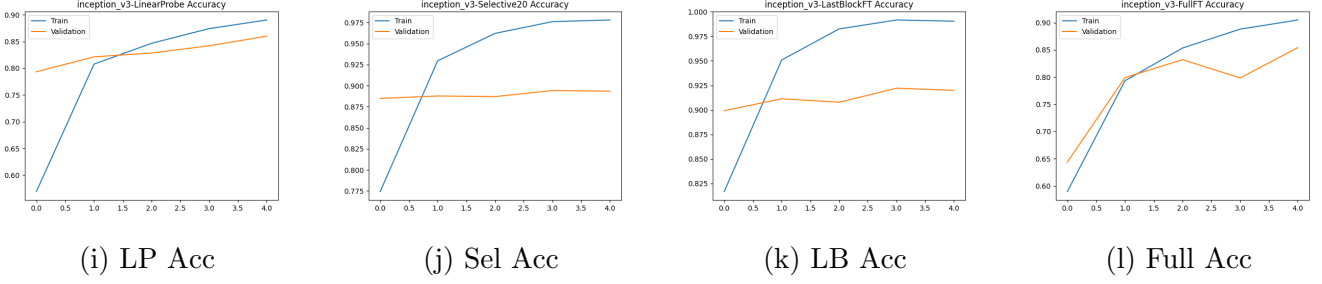


Figure 7: Inception v3: Training Accuracy across strategies (Continued).

4.6 Gradient Statistics and Convergence Stability

- **Gradient Behavior:** Deeper layers exhibit higher gradient norms, confirming they undergo the most significant adaptation. **Inception v3** experienced a massive initial gradient norm spike (1001.76) during Full FT, signaling optimization instability that eventually led to a performance collapse (85.4%).
- **Convergence Stability:** **EfficientNet-B0** and **ResNet50** show smoother loss decay in Selective and Last Block modes. Full FT often introduces "noise" into the loss curves, as the model struggles to balance updating low-level filters with high-level semantics simultaneously.

4.7 Layer Adaptation and Representation Quality

- **Early vs. Deep Layers:** Early layers provide universal features (edges, color gradients). Modifying them via Full FT tends to degrade generalization as it overwrites robust ImageNet knowledge. Deeper layers capture task-specific geometry.
- **Selective Reasoning:** We unfroze the final **20% of parameters** because high-level semantic concepts are concentrated in the tail of the backbone. Freezing the early 80% acts as a regularizer, ensuring the model retains its "visual vocabulary" while specializing its "reasoning" for the AID dataset.

4.8 Conditions of Reduced Robustness

- **Failure Condition:** Robustness is severely compromised when unfreezing complex, high-capacity models on datasets that lack the scale to constrain the expanded parameter space. This is most evident in **Inception v3**, where Full FT resulted in a validation accuracy of only **85.4%**, significantly lower than its **92.0%** peak during Last Block FT.
- **Mechanism (Catastrophic Forgetting):** The underlying cause is the destruction of the pre-trained feature manifold. In **Inception v3**, the massive initial **Gradient Norm of 1001.76** during Full FT indicates that backpropagation updates were large enough to overwrite robust, generalized ImageNet weights.
- **Data-Backed Evidence:**
 - **ResNet50:** Full FT (100% parameters) achieved **93.4%**, a slight drop from Last Block FT (63.49% parameters) at **93.9%**, showing the onset of over-specialization.
 - **Generalization Gap:** In **EfficientNet-B0**, while Full FT reached **93.1%**, the Grad-Norm started at a high **662.82** compared to just **125.08** for Last Block FT, suggesting that full unfreezing introduces optimization noise that can hinder out-of-distribution performance.

5 Few-Shot Learning Analysis

In this scenario, we evaluate the data efficiency of each backbone under limited supervision by training models using subsets of 100%, 20%, and 5% of the training data.

5.1 Quantitative Results

The table below summarizes the performance metrics and overfitting severity across different data regimes.

Model Architecture	100% Data	20% Data	5% Data	Relative Drop	Train-Val Gap @ 5%
ResNet50	0.9321	0.8385	0.7327	0.2140	0.2637 (High)
EfficientNet-B0	0.9314	0.8528	0.7062	0.2417	0.2902 (High)
Inception v3	0.9099	0.8256	0.6104	0.3291	0.2820 (High)

Table 4: Few-Shot Learning Performance Summary

Model Architecture	Gauss 0.05	Gauss 0.1	Gauss 0.2	Motion Blur	Brightness	Avg Corruption Error
ResNet50	0.0722	0.1229	0.3031	0.4446	0.1480	0.2181
EfficientNet-B0	0.1001	0.2473	0.8320	0.4246	0.0951	0.3398
Inception v3	0.1601	0.2788	0.6254	0.6483	0.1851	0.3795

Table 5: Corruption Error ($1 - Acc_{corrupted}$) under various distribution shifts. Lower values indicate higher robustness.

5.2 Visualizations

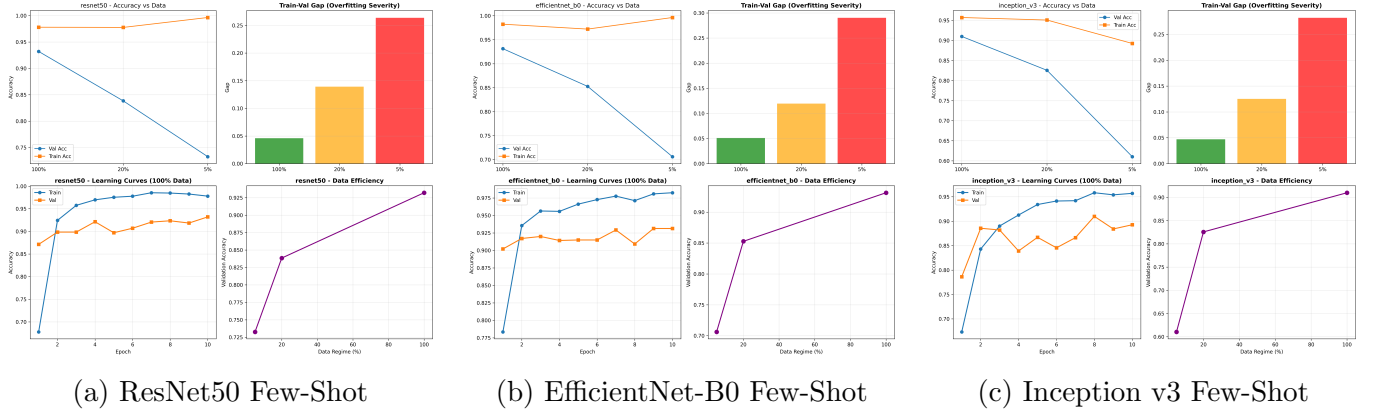


Figure 8: Performance degradation and training-validation gap analysis under limited data regimes.

5.3 Analytical Discussion

- **Data Efficiency: ResNet50** demonstrates the strongest data efficiency, retaining the highest accuracy (73.27%) at the 5% data regime and experiencing the lowest relative performance drop (21.40%). Its residual connections likely act as an implicit ensemble of shallower networks, preventing the model from drastically overfitting to a small number of samples.
- **Model Capacity vs. Overfitting:** All models experience severe overfitting at the 5% regime, with train-validation gaps exceeding 0.26. **Inception v3** performs the poorest under limited data (61.04% accuracy), suffering a massive 32.91% relative drop. Its complex parallel filter architecture (Inception modules) requires substantial data to properly optimize the concatenated feature maps without memorizing the sparse training set.

6 Corruption Robustness Evaluation

In this scenario, we analyze robustness under controlled distribution shifts applied at evaluation time: Gaussian noise ($\sigma = 0.05, 0.1, 0.2$), motion blur, and brightness shift.

6.1 Quantitative Results

Model Architecture	Clean Acc	Gauss 0.05	Gauss 0.1	Gauss 0.2	Motion Blur	Brightness	Avg Rel Robustness
ResNet50	0.9342	0.9278	0.8771	0.6969	0.5554	0.8520	0.8369
EfficientNet-B0	0.9307	0.8999	0.7527	0.1680	0.5754	0.9049	0.7094
Inception v3	0.8542	0.8399	0.7212	0.3746	0.3517	0.8149	0.7264

Table 6: Corruption Robustness Performance

6.2 Visualizations

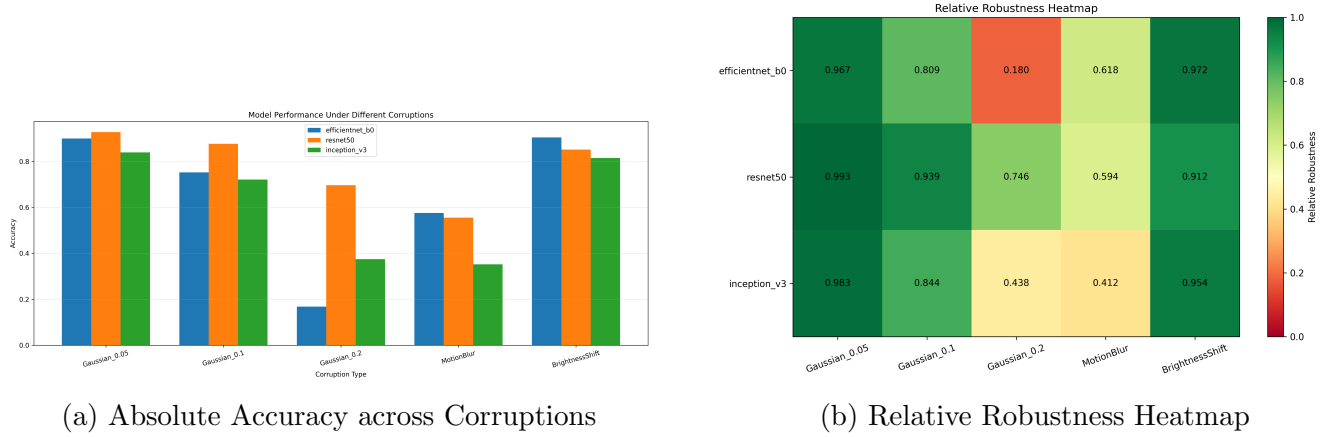


Figure 9: Comparison of model degradation under various distribution shifts.

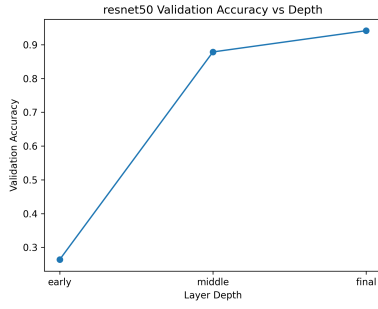
6.3 Analytical Discussion

- **Noise Robustness:** ResNet50 is overwhelmingly the most robust to high-frequency pixel corruption, maintaining $\sim 69.7\%$ accuracy even under severe Gaussian noise ($\sigma = 0.2$). The skip connections allow the network to bypass corrupted localized features and rely on broader, uncorrupted spatial representations.
- **Architectural Sensitivity:** EfficientNet-B0 is highly sensitive to Gaussian noise, suffering a catastrophic drop to 16.8% accuracy at $\sigma = 0.2$. However, it is the most robust model against spatial and illumination shifts, achieving the highest accuracy for Motion Blur (57.5%) and Brightness Shift (90.5%). Its use of Depthwise Separable Convolutions makes it highly efficient but apparently fragile when individual pixel values are randomly perturbed.

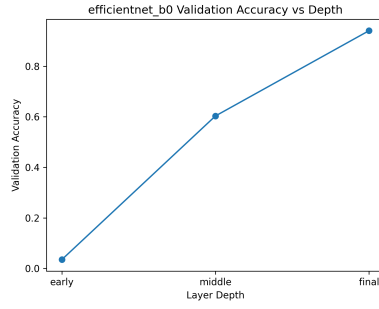
7 Layer-Wise Feature Probing

To examine how semantic abstraction evolves, we extracted intermediate representations from early, middle, and final layers of the models and trained separate linear classifiers. As strictly required by the experimental setup, all PCA visualizations across different layers and models were generated using a consistently fixed subset of 900 images (30 classes, 30 samples per class) to ensure a fair comparison of the feature spaces.

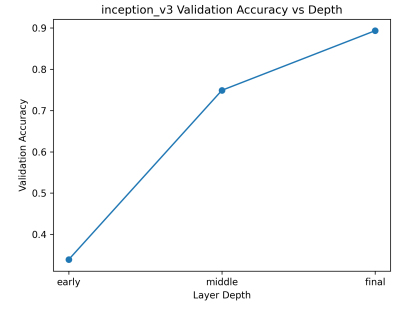
7.1 Visualizations



(a) ResNet50 Probe Acc

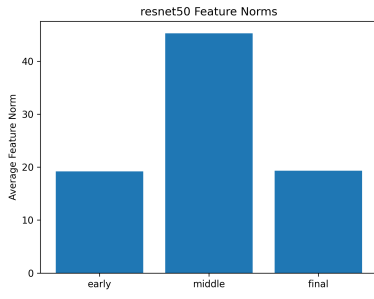


(b) EfficientNet-B0 Probe Acc

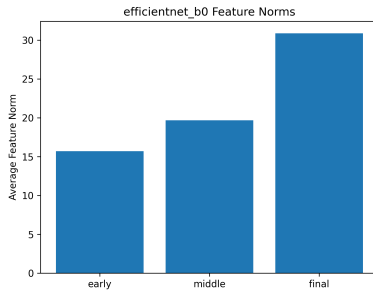


(c) Inception v3 Probe Acc

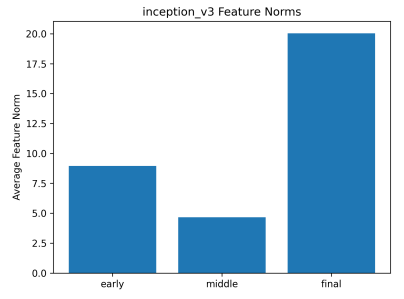
Figure 10: Validation Accuracy vs. Model Depth for all architectures.



(a) ResNet50 Feature Norms

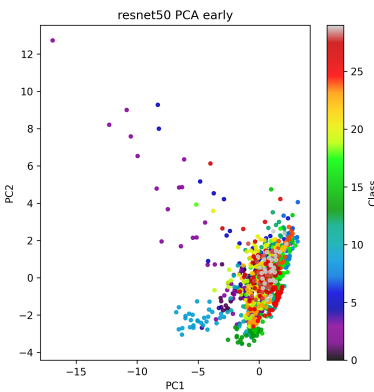


(b) EfficientNet-B0 Feature Norms

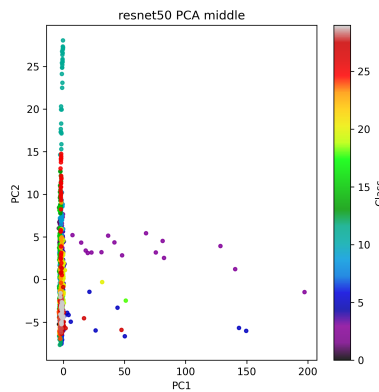


(c) Inception v3 Feature Norms

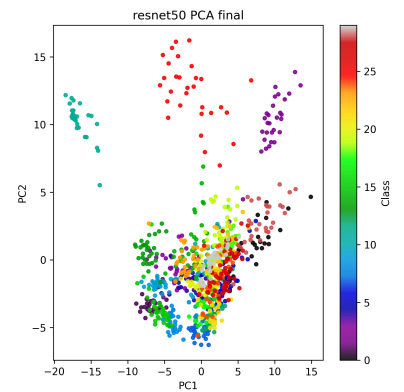
Figure 11: Feature Norm Statistics across selected layers (Early, Middle, Final).



(a) ResNet50 Early



(b) ResNet50 Middle



(c) ResNet50 Final

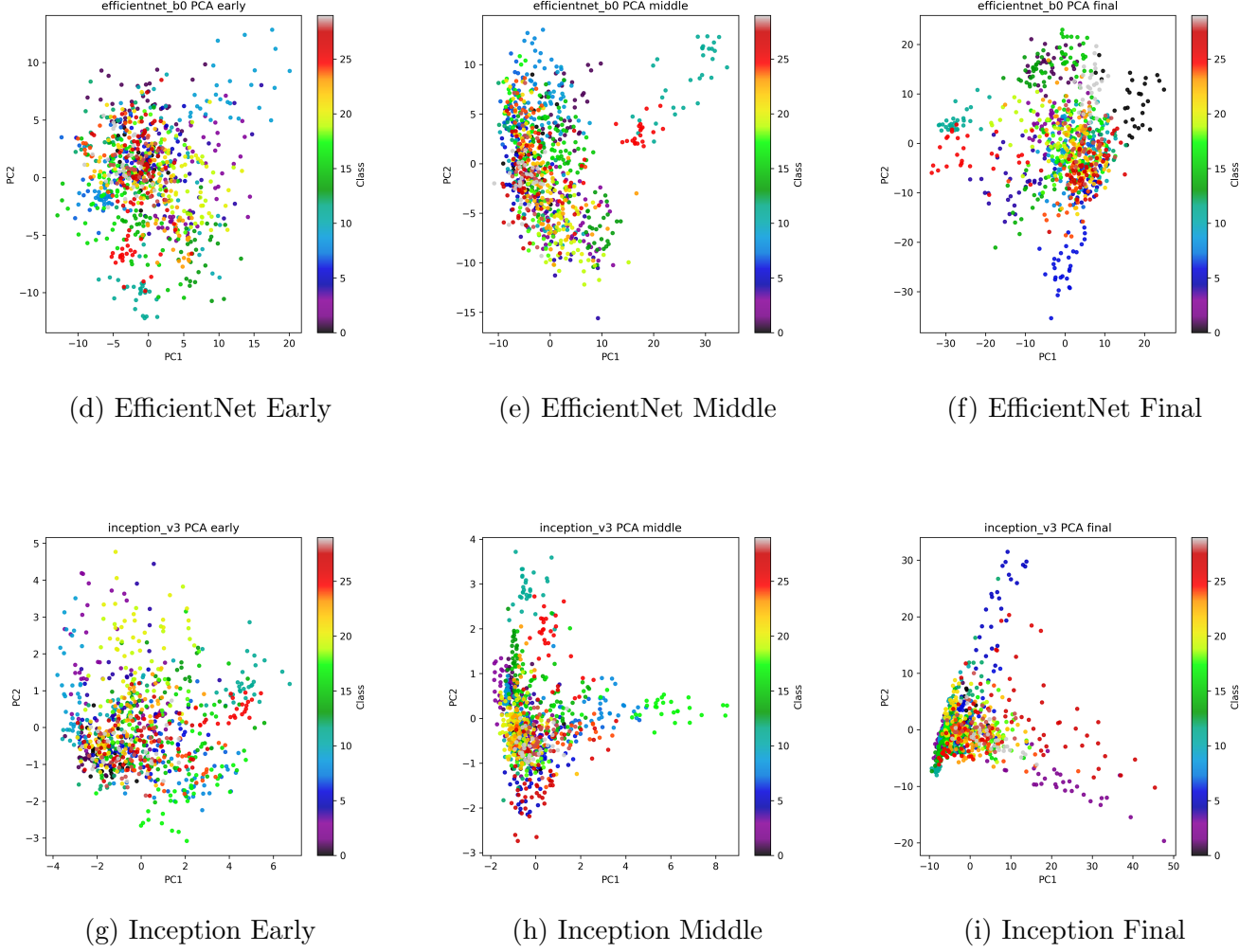


Figure 12: Evolution of PCA feature space across network depth. Clusters become tighter and more class-separable as depth increases.

7.2 Analytical Discussion

- **Semantic Abstraction Evolution:** The PCA plots demonstrate a clear evolution of semantic abstraction. Early layer representations show massive overlap across all 30 classes, indicating that early filters are responding to generic, low-level features (edges, color blobs) rather than class-specific geometries. By the final layers, the feature space unrolls, and distinct, compact clusters form, matching the high validation accuracy of the deep probes.
- **Hierarchy and Transferability:** The middle layers show partial clustering. This hierarchy proves that transferability is layer-dependent: early layers transfer well universally across vastly different domains, but final layers encode the highly specific spatial semantics required to differentiate between complex aerial classes (e.g., distinguishing between different densities of residential areas).
- **Feature Norm Evolution:** The feature norm statistics demonstrate noticeable shifts as representations transition from low-level generic features to high-level semantic abstractions. Changes in feature magnitudes at deeper layers correlate with the network’s increasing confidence and class separability, which directly supports the improved linear probing accuracy.

8 Key Insights & Intellectual Expectations

8.1 Unexpected Finding

We observed a catastrophic failure in **EfficientNet-B0** under severe Gaussian noise ($\sigma = 0.2$), where its accuracy dropped dramatically to **16.8%**. Surprisingly, this same model maintained excellent robustness against motion blur and brightness shifts. This highlights a specific vulnerability of **Depthwise Separable Convolutions** to high-frequency, uncorrelated pixel noise compared to standard convolutions found in ResNet50.

8.2 Addressing Core Intellectual Expectations

- **Which architecture transfers best and why?**

Inception v3 transfers best out-of-the-box as a frozen feature extractor (Linear Probe Acc: 86.0%). Its multi-branch architecture with varying filter sizes (1×1 , 3×3 , 5×5) allows it to inherently capture objects at multiple scales, which is highly advantageous for aerial imagery where object resolutions vary drastically.

- **Which model is most robust under distribution shift?**

ResNet50 proved to be the most robust overall, yielding an average relative robustness of **0.8369**. Its residual skip connections provide alternative pathway routing that mitigates the destructive effect of localized pixel corruptions.

- **Which layers encode the most transferable semantics?**

The **early-to-middle layers** encode the most universally transferable semantics (textures, boundaries), but for domain-specific tasks like aerial classification, the **final 20% of layers** encode the spatial hierarchies (e.g., repeating housing patterns) necessary for high accuracy.

- **When does fine-tuning degrade robustness?**

Fine-tuning degrades robustness when complex, high-capacity models (like Inception v3) are **fully unfrozen (Full FT)** on datasets that lack the scale to constrain the expanded parameter space. This causes "Catastrophic Forgetting," wiping out the generalized ImageNet manifold and resulting in overfitting (evidenced by massive gradient norm spikes and lower validation performance compared to Selective unfreezing).

- **How does inductive bias manifest empirically?**

Inductive bias manifests clearly in the **efficiency metrics**. EfficientNet-B0 achieves competitive clean fine-tuned accuracy (93.1%) using only **4.00M parameters and 0.77 GFLOPs**, compared to ResNet50's **23.57M parameters and 8.26 GFLOPs**. The compound scaling inductive bias allows EfficientNet to encode highly discriminative features in a fraction of the computational footprint.